

Lab 1: UNIX/Linux System Administration

CNIT 34000

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Procedures

Creating and Installing Rocky Linux

- Accessed Nutanix cluster by navigating to <http://virtual-lab.cit.lcl> and logging in via Boilerad
- Created new Rocky Linux virtual machine in VM tab under Computing
 - Named the VM CNIT340.G171.Rocky
 - Assigned 4 GB of RAM
 - Enabled Secure Boot and vTPM
- Added 60 GB hard disk with SCSI bus type
- Added CD-ROM with the Rocky Linux 9.8 SATA image
- Configured network adapter to use CNIT340 Linux A network
- Set the time zone to UTC with the region configured as Indiana/Indianapolis
- Saved VM changes
- Powered on the VM and booted
- Selected minimal install
- Configured Installation Destination and partitions
 - Selected the 60 GB disk
 - Chose Custom storage configuration
 - Set /home to 4096 MiB ext4 LVM
 - Set / to 45903 MiB ext4 LVM
 - Set /boot/efi to 600 MiB efi Standard Partition
 - Set swap to 6144 MiB swap LVM
 - Set /boot to 2000 MiB ext4 Standard Partition

- Set /var to 4096 MiB ext4 LVM
- Configured network and hostname
 - Entered btrimbl.rocky as the hostname
 - Enabled the ethernet adapter
 - Configured static IPv4 settings: address, subnet mask, gateway, and DNS servers
- Created a root password and an administrative user account
 - Entered name Brendan Trimble and username btrimbl
 - Checked “Make this user administrator.”
- Finished installation and rebooted VM
- Clicked Unmount on Rocky Linux CD-ROM
- Disabled SELinux
 - Opened /etc/selinux/config with `sudo nano /etc/selinux/config`
 - Changed SELinux=enforcing to SELinux=disabled
 - Saved and rebooted
 - Verified with `getenforce` and confirmed it as Disabled

Creating and Installing Windows Server 2022

- Created Windows Server VM in Nutanix named CNIT340.G171.Trimble.Windows with 8 GB of RAM
- Added a 48 GB hard disk
- Added CD-ROM Windows Server 2022 SATA image
- Configured adapter with CNIT340WL Windows network
- Enabled Memory Overcommit, Secure Boot, and Attach vTPM
- Powered on VM and booted

- Selected standard Windows Server installation option with Desktop Experience
- Completed installation and rebooted the VM
- Clicked Unmount on Windows Server 2022 CD-ROM
- Enabled Ethernet connectivity
 - Installed netkvm driver in File Explorer under Nutanix VirtIO \ Windows Server 2022 \ x64
 - Ran ncpa.cpl from the terminal
- Configured static IPv4 address, subnet mask, gateway, and DNS servers in settings under Network & Internet
 - Set Ethernet to manual
 - Set IPv4 address to 44.101.0.171
 - Set IPv4 mask to 255.255.254.0
 - Set IPv4 gateway to 44.101.0.1
 - Set IPv4 DNS Servers to 44.2.1.44 & 44.2.1.45
- Updated Windows Server using Windows Update
- Installed all available updates and restarted as required

Creating and Installing Ubuntu

- Created Ubuntu VM in Nutanix named CNIT340.G171.Trimble.Ubuntu with 4 GB of RAM
- Added 60 GB hard disk using SCSI bus type
- Added CD-ROM with Ubuntu 24.04.4 SATA image
- Configured adapter to use CNIT340 Linux B network
- Enabled Memory Overcommit, Secure Boot, and Attach vTPM

- Configured time zone to UTC
- Set language to English
- Configured network
 - Set subnet to 44.101.4.0/23
 - Set gateway to 44.101.4.1
 - Set address to 44.101.4.171
 - Set time servers to 44.2.1.44 and 44.2.1.45
- Formatted free space during installation
 - Set partition 1 as /boot ext4 (4GB)
 - Set partition 2 as /home ext4 (4GB)
 - Set partition 3 as /var ext4 (4GB)
 - Set partition 4 as / ext4 (46.948GB)
- Logged into Ubuntu Linux with boilerad credentials
- Powered on the VM and booted without a GUI
- Logged into Ubuntu server with created account
- Enabled and started SSH
 - Entered systemctl enable ssh
 - Typed systemctl start ssh
- Verified NFS was not installed or running with systemctl status nfs-server
- Verified Bash was installed with bash --version
- Configured a 6 GB swap file
 - Checked existing swap with swapon --show
 - Created the file with sudo fallocate -l 6G /swapfile

- Secured it with `sudo chmod 600 /swapfile`
- Formatted it with `sudo mkswap /swapfile`
- Activated it with `sudo swapon /swapfile`
- Configured swap file to stay after reboot
 - Added `/swapfile none swap sw 0 0` to `/etc/fstab`
 - Verified with `grep swap /etc/fstab`

Installing NGT Tools on Rocky Linux

- Mounted Nutanix Guest Tools from the Summary tab in Nutanix by selecting More | Guest Tools | Set Up NGT
- Located mount in Linux console with `blkid -L NUTANIX_TOOLS`
- Mounted the NGT ISO with `sudo mount /dev/sr0 /mnt`
- Installed NGT with `sudo python /mnt/installer/linux/install_ngt.py --operation install`

Installing NGT Tools on Ubuntu

- Mounted Nutanix Guest Tools from the Summary tab in Nutanix
- Located the mounted device in the Ubuntu console with `blkid -L NUTANIX_TOOLS`
- Mounted the NGT ISO with `sudo mount /dev/sr0 /mnt`
- Installed NGT with `sudo python /mnt/installer/linux/install_ngt.py --operation install`
- Waited for the installation to complete and confirmed the `ngt_guest_agent` service was active and running

Installing NGT Tools on Windows Server 2022

- Mounted NGT from the Summary tab in Nutanix by selecting More | Guest Tools | Set Up NGT

- Returned to the Windows Server session and opened the mounted NGT drive in File Explorer
- Ran Setup and followed the installation prompts
- Completed the installation successfully

Configuring Rocky Linux

- Logged into Rocky Linux and verified the hostname
- Verified the running IP configuration with `ip addr` and confirmed it matched the assigned IP Address Sheet
- Verified the default router with `ip route`
- Verified DNS servers with `nmcli device show`
- Verified the boot-time network configuration with `nmcli connection show` and confirmed it matched the IP Address Sheet
- Tested internet connectivity with `ping -c 4 8.8.8.8` and confirmed successful replies
- Verified Bash was installed with `bash --version`

Configuring Ubuntu

- Logged into Ubuntu and verified the hostname with the `hostname` command
- Verified the running IP configuration with `ip addr` and confirmed it matched the assigned IP Address Sheet
- Verified the default router with `ip route`
- Verified DNS with `resolvectl status`
- Verified the boot-time configuration by reviewing `/etc/netplan/01-netcfg.yaml` with `cat` and confirmed it matched the IP Address Sheet

- Tested internet connectivity with `ping -c 4 8.8.8.8` and confirmed successful replies
- Verified Bash was installed with `bash --version`
- Verified SSH was active with `systemctl status ssh`
- Verified the NFS server was not enabled with `systemctl status nfs-server`

Creating User Accounts on Rocky Linux

- Created the jack account with `sudo useradd -m jack` and assigned a password with `sudo passwd`
- Created the jill account with `sudo useradd -m jill` and assigned a password with `sudo passwd`
- Verified both accounts existed with `getent passwd jack jill`
- Verified home directories with `ls -l /home`, confirming `/home/jack` and `/home/jill` existed
- Verified home directory permissions with `ls -ld /home/jack /home/jill` and confirmed they were appropriate

Creating User Accounts on Ubuntu

- Created the jack account with `sudo useradd -m -s /bin/bash jack` and assigned a password with `sudo passwd`
- Created the jill account with `sudo useradd -m -s /bin/bash jill` and assigned a password with `sudo passwd`
- Verified both accounts existed with `getent passwd jack jill`
- Verified home directories with `ls -l /home`
- Verified home directory permissions with `ls -ld /home/jack /home/jill` and confirmed they were correct

Configuring Password Policies on Rocky Linux

- Opened `/etc/security/pwquality.conf` with `sudo vi /etc/security/pwquality.conf`
- Set the minimum password length with `minlen = 9`
- Configured mixed case with `ucrcedit = -1` and `lcredit = -1`
- Tested the policy
 - Attempted to assign Jack a password that did not meet the requirements and confirmed it was rejected
 - Assigned a password that met the requirements and confirmed it was accepted

Configuring Password Policies on Ubuntu

- Installed `libpam-pwquality` with `sudo apt install libpam-pwquality`
- Opened `/etc/security/pwquality.conf` with `sudo vi /etc/security/pwquality.conf`
- Set values for Rocky Linux to `minlen = 9`, `ucrcedit = -1`, and `lcredit = -1`
- Tested the policy the same way, confirming a noncompliant password was rejected and a compliant password was accepted

Configuring Quotas on Rocky Linux

- Installed the quota package with `sudo dnf install quota -y`
- Edited `/etc/fstab` with `sudo vi /etc/fstab` to add `usrquota` and `grpquota` to the `/home` entry
- Remounted `/home` and initialized quotas with `sudo mount -o remount /home && sudo quotacheck -cum /home && sudo quotaon /home`
- Configured quotas for jack and jill with `sudo edquota -u jack` and `sudo edquota -u jill`
- Verified the assigned limits with `sudo repquota /home`

Configuring Quotas on Ubuntu

- Installed the quota package with `sudo apt update && sudo apt install quota -y`
- Edited `/etc/fstab` with `sudo vi /etc/fstab` to add `usrquota` and `grpquota` to the `/home` entry
- Remounted `/home` and initialized quotas with `sudo mount -o remount /home && sudo quotacheck -cum /home && sudo quotaon /home`
- Configured quotas for jack and jill with `sudo edquota -u jack` and `sudo edquota -u jill`
- Verified the assigned limits with `sudo repquota /home`

Configuring Printing on Rocky Linux

- Used the printer address `44.44.113.254` for WTHR 313
- Installed CUPS with `sudo dnf install cups -y`
- Enabled and started CUPS with `sudo systemctl enable --now cups` and confirmed it was active
- Added the WTHR 313 network printer and set it as the default with `sudo lpadmin -p LabPrinter -E -v socket://44.44.113.254 -m raw`
- Tested command-line text printing
 - Created a text file named `printtest.txt`
 - Printed it with `lp -d LabPrinter printtest.txt` and confirmed it printed successfully without stair-stepping
- Tested graphics printing
 - Created a graphics file with `convert -size 500x200 xc:white -fill black -pointsize 30 -gravity center -annotate 0 "CNIT 340 Rocky Linux" printtest.png`
 - Printed it with `lp -d LabPrinter printtest.png` and confirmed it printed successfully

Configuring Printing on Ubuntu

- Used the printer address 44.44.113.254 for WTHR 313
- Installed CUPS and the CUPS client with `sudo apt update` followed by `sudo apt install cups cups-client -y`
- Enabled and started CUPS with `sudo systemctl enable --now cups` and confirmed it was active
- Added the WTHR 313 network printer and set it as the default with `sudo lpadmin -p LabPrinter -E -v socket://44.44.113.254 -m raw`
- Tested command-line text printing with `lp -d LabPrinter printtest.txt` and confirmed success without stair-stepping
- Tested graphics printing
 - Installed ImageMagick with `sudo apt install imagemagick -y`
 - Created a graphics file with `convert -size 500x200 xc:white -fill black -pointsize 30 -gravity center -annotate 0 "CNIT 340 Ubuntu" printtest.png`
 - Printed it with `lp -d LabPrinter printtest.png` and confirmed it printed successfully

Patching Rocky Linux

- Updated the system with `sudo dnf upgrade -y` and confirmed it completed successfully
- Checked for remaining updates with `sudo dnf check-update` and confirmed none of significance remained

Patching Ubuntu

- Updated the system with `sudo apt update && sudo apt upgrade -y` and confirmed it completed successfully

- Checked for remaining updates with apt list --upgradable and confirmed none of significance remained

Linux OS Comparison

Similarities:

- Both Rocky and Ubuntu were installed without a Graphical User Interface (GUI)
- Both Rocky and Ubuntu allowed configurations for IPv4 address, subnet mask, gateway, and DNS servers during installation
- Both Rocky and Ubuntu required separate /home and /var partitions
- The /home directory was configured on both Linux machines with two users (jack and jill) and with both permissions and quotas for those users
- Both operating systems required password requirements for users
- CUPS network printing was enabled on both operating systems with the ability to print test pages to WTHR 313
- NGT was mounted and configured on both Linux machines
- Both required BASH to be installed

Differences:

- Rocky Linux required SELinux to be disabled, while it was not needed for Ubuntu
- Ubuntu and Rocky Linux had their respective drives partitioned differently
 - Ubuntu required 4GB ext4 for /home, /var and /boot with the rest of the storage on /
 - Rocky required different partitions for /boot/efi and the swap partition with different partitions requiring either standard or LVM

Appendix A: Screenshots

```
jack@btrimbl:~$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                                  8:0      0   60G  0 disk
├─sda1                              8:1      0  600M  0 part /boot/efi
├─sda2                              8:2      0    2G  0 part /boot
└─sda3                              8:3      0  57.5G  0 part
   ├─rl--root-root                 253:0      0  44.8G  0 lvm /
   ├─rl--root-swap                 253:1      0    6G  0 lvm [SWAP]
   ├─rl--root-home                 253:2      0    4G  0 lvm /home
   └─rl--root-var                 253:3      0   2.6G  0 lvm /var
sr0                                  11:0      1 1024M  0 rom
sr1                                  11:1      1   16M  0 rom
jack@btrimbl:~$
```

Figure 1) Disk partition configuration for Rocky Linux VM

```
jack@btrimbl:~$ sudo repquota /home
*** Report for user quotas on device /dev/mapper/rl--root-home
Block grace time: 7days; Inode grace time: 7days
Block limits
File limits
User      used    soft    hard    grace    used    soft    hard    grace
-----
root      --      20      0      0      --      2      0      0
jack      --    8188 102400 122880      --    507      0      0
jill      --      28 102400 122880      --      7      0      0
jack@btrimbl:~$
```

Figure 2) User quotas configured for jack and jill in home directory of Rocky Linux

```
GNU nano 8.1 /etc/security/pwquality.conf Modified
# Number of characters in the new password that must not be present in the
# old password.
# difok = 1
#
# Minimum acceptable size for the new password (plus one if
# credits are not disabled which is the default). (See pam_cracklib manual.)
# Cannot be set to lower value than 6.
# minlen = 9
#
# The maximum credit for having digits in the new password. If less than 0
# it is the minimum number of digits in the new password.
# dcredit = 0
#
# The maximum credit for having uppercase characters in the new password.
# If less than 0 it is the minimum number of uppercase characters in the new
# password.
# ucredit = -1
#
# The maximum credit for having lowercase characters in the new password.
# If less than 0 it is the minimum number of lowercase characters in the new
# password.
# lcredit = -1
#
# The maximum credit for having other characters in the new password.
# If less than 0 it is the minimum number of other characters in the new
# password.
# ocredit = 0
#
# The minimum number of required classes of characters for the new
# password (digits, uppercase, lowercase, others).
# minclass = 0
#
# The maximum number of allowed consecutive same characters in the new password.
# The check is disabled if the value is 0.
# maxrepeat = 0
#
# The maximum number of allowed consecutive characters of the same class in the
# new password.
# The check is disabled if the value is 0.
# maxclassrepeat = 0
#
# Whether to check for the words from the passwd entry GECOS string of the user.
# The check is enabled if the value is not 0.
# geccoscheck = 0
#
# Whether to check for the words from the cracklib dictionary.
#
# Help      Write Out  Where Is  Cut       Execute   Location  Undo      Set Mark  To Bracket Previous
# Exit      Read File  Replace  Paste     Justify   Go To Line Redo      Copy      Where Was  Next
```

Figure 3) User password configuration file for Rocky Linux

```
jack@btrimbl:~$ lpstat -v NetworkPrinter
device for NetworkPrinter: socket://44.44.113.254
jack@btrimbl:~$ _
```

Figure 4) Default printer called NetworkPrinter set to WTHR 313 printer IP on Rocky Linux

```

jack@btrimbl:~$ sudo cat /etc/cups/printers.conf
# Printer configuration file for CUPS v2.4.10
# Written by cupsd
# DO NOT EDIT THIS FILE WHEN CUPSD IS RUNNING
NextPrinterId 3
<Printer NetworkPrinter>
PrinterId 2
UUID urn:uuid:1bf94ae5-da11-3df8-50de-a9435ee69adf
Info NetworkPrinter
MakeModel Generic PostScript Printer
DeviceURI socket://44.44.113.254
State Idle
StateTime 1789563787
ConfigTime 1789522299
Type 8400972
Accepting Yes
Shared Yes
JobSheets none none
QuotaPeriod 0
PageLimit 0
KLimit 0
OpPolicy default
ErrorPolicy stop-printer
Attribute marker-colors \#000000
Attribute marker-levels 25
Attribute marker-names Black Cartridge HP CF226X
Attribute marker-types toner
Attribute marker-change-time 1789563787
</Printer>
jack@btrimbl:~$

```

Figure 5) CUPS printer configuration file for WTHR 313 printer on Rocky Linux

```

btrimbl@btrimbl:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0   60G  0 disk
├─sda1       8:1    0    1G  0 part /boot/efi
├─sda2       8:2    0    4G  0 part /boot
├─sda3       8:3    0    4G  0 part /home
├─sda4       8:4    0    4G  0 part /var
└─sda5       8:5    0  46.9G  0 part /
sr0         11:0    1   387M  1 rom
sr1         11:1    1  1024M  1 rom
btrimbl@btrimbl:~$

```

Figure 6) Disk partition configuration for Ubuntu Linux VM

```

btrimbl@btrimbl:~$ sudo repquota /home
*** Report for user quotas on device /dev/sda3
Block grace time: 7days; Inode grace time: 7days

```

Block limits					File limits				
User		used	soft	hard	grace	used	soft	hard	grace
root	--	20	0	0		2	0	0	
btrimbl	--	28	0	0		10	0	0	
jack	--	32	102400	122880		12	0	0	
jill	--	20	102400	122880		5	0	0	

```

btrimbl@btrimbl:~$

```

Figure 7) User quotas configured for jack and jill in the home directory of Ubuntu Linux

```

GNU nano 7.2 /etc/security/pwquality.conf
# Configuration for systemwide password quality limits
# Defaults:
#
# Number of characters in the new password that must not be present in the
# old password.
# difok = 1
#
# Minimum acceptable size for the new password (plus one if
# credits are not disabled which is the default). (See pam_cracklib manual.)
# Cannot be set to lower value than 6.
# minlen = 9
#
# The maximum credit for having digits in the new password. If less than 0
# it is the minimum number of digits in the new password.
# dcredit = 0
#
# The maximum credit for having uppercase characters in the new password.
# If less than 0 it is the minimum number of uppercase characters in the new
# password.
# ucredit = -1
#
# The maximum credit for having lowercase characters in the new password.
# If less than 0 it is the minimum number of lowercase characters in the new
# password.
# lcredit = -1
#
# The maximum credit for having other characters in the new password.
# If less than 0 it is the minimum number of other characters in the new
# password.
# ocredit = 0
#
# The minimum number of required classes of characters for the new
# password (digits, uppercase, lowercase, others).
# minclass = 0
#
# The maximum number of allowed consecutive same characters in the new password.
# The check is disabled if the value is 0.
# maxrepeat = 0
#
# The maximum number of allowed consecutive characters of the same class in the
# new password.
# The check is disabled if the value is 0.
# maxclassrepeat = 0
#
# Whether to check for the words from the passwd entry GECOS string of the user.
# The check is enabled if the value is not 0.
#
[ Read 79 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^-A Set Mark  ^-_ To Bracket ^_O Previous
^X Exit      ^R Read File  ^N Replace    ^P Paste      ^_ Justify    ^_ Go To Line ^E Redo       ^-S Copy      ^_ Where Was  ^_N Next

```

Figure 8) User password configuration file for Ubuntu Linux

```

btrimbl@btrimbl:~$ lpstat -v NetworkPrinter
device for NetworkPrinter: socket://44.44.113.254
btrimbl@btrimbl:~$

```

Figure 9) Default printer called NetworkPrinter set to WTHR 313 printer IP on Ubuntu Linux

```

Option cups-browsed true
Option print-color-mode monochrome
</Printer>
<Printer DUDL_Printer_sherwoob_ubt>
PrinterId 61
UUID urn:uuid:e330c038-0775-37eb-694e-f00553d2c0a9
Info DUDL_Printer
Location
MakeModel LaserJet Printer, driverless, 2.0.0
DeviceURI implicitclass://DUDL_Printer_sherwoob_ubt/
State Idle
StateTime 1790035218
ConfigTime 1790035218
Type 36958
Accepting Yes
Shared No
JobSheets none none
QuotaPeriod 0
PageLimit 0
KLimit 0
OpPolicy default
ErrorPolicy retry-job
Option cups-browsed true
Option print-color-mode monochrome
</Printer>
<Printer NetworkPrinter>
PrinterId 11
UUID urn:uuid:420b0497-1168-3f59-42c5-d4d3ddc8dca8
Info NetworkPrinter
MakeModel Generic PostScript Printer
DeviceURI socket://44.44.113.254
State Idle
StateTime 1789559999
ConfigTime 1789558994
Type 8400972
Accepting Yes
Shared Yes
JobSheets none none
QuotaPeriod 0
PageLimit 0
KLimit 0
OpPolicy default
ErrorPolicy retry-job
Attribute marker-colors \#000000
Attribute marker-levels 25
Attribute marker-names Black Cartridge HP CF226X
Attribute marker-types toner
Attribute marker-change-time 1789559999
</Printer>
btrimbl@btrimbl:~$

```

Figure 10) CUPS printer configuration file for WTHR 313 printer on Ubuntu Linux

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